Biochemical Engineering Journal 214 (2025) 109585

Contents lists available at ScienceDirect

Biochemical Engineering Journal

journal homepage: www.elsevier.com/locate/bej

Regular article

CFD predictive simulations of miniature bioreactor mixing dynamics coupled with photo-bioreaction kinetics in transitional flow regime

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ARTICLE INFO

Keywords:
Mechanical mixing
Miniaturised stirred bioreactors
CFD modelling
Photobioreactor
Bioreaction kinetics
Light attenuation and transport

ABSTRACT

High-throughput systems using miniaturised stirred bioreactors accelerate bioprocess development due to their simplicity and low cost. However, fluctuating hydrodynamics pose numerical challenges for coupling (bio)reaction kinetics, critical for optimisation and scale-up/down in chemical and bioprocess industries. To address this, hydrodynamic convergence was achieved by time-averaging instantaneous RANS solutions of the *transitional* SST model over a sufficiently long period to achieve statistical significance in step one. Subsequently, photo-bioreaction transport models, accounting for the photobioreactor's directional illumination and curvature, were solved based on these converged fields, overcoming two-step coupling challenges in an approach not previously reported. Applied to a 0.7 L Schott bottle photobioreactor mechanically mixed by a magnetic stirrer (100–500 rpm), the model accurately predicted swirly vortex fields at 500 rpm, with a 7 % error margin for simulated tracer diffusion, and aligned biomass growth profiles with literature data on *Rhodopseudomonas palustris*. However, parallel computing efficiency did not scale linearly with processor count, making time-averaging computationally expensive. Also, improved bioreactor mixing enhanced biomass productivity, but rpms over 300 required increased incident light intensity (>100 Wm^{-2}) due to observed light limitation. Hence, this model facilitates optimising stirring speeds and refining operational parameters for scale-up and scale-down processes.

1. Introduction

As the world's population increases rapidly, the urge to produce sustainable bioproducts increases, leading to the need to understand the physical structures required to enable large-scale production. Examples of such structures are biological reactors (bioreactors) which provide suitable conditions for microbial cells to bio-convert substrates into desired bioproducts (such as biochemicals, biofuels, and biomaterials). However, various factors influence their performance, with mixing being a crucial engineering variable that has been extensively studied in the literature featuring mechanical, pneumatic and natural circulation techniques. Mechanical mixing can be achieved internally, as in stirred tank reactors [1–4], magnetic stirrers [5,6] and raceway ponds [7,8], or externally, as in Orbital shakers [9,10] and rocking motions [11,12]. Pneumatic methods involve pumps or gas sparging [13–15] while natural circulation relies on temperature-induced density differences for buoyancy-driven motion [16,17].

In the chemical and bioprocess industries, system homogeneity and mass transfer are vital. Hence, mechanical mixing methods dominate throughout the different stages of process development, guided by essential scale-up and scale-down criteria such as constant power per unit volume [9,18–22], constant mixing time [18,20,22], constant Reynolds number [18,19], geometrical similarity [1,23] and constant impeller tip speed [18,19,23]. Also, they could be combined with pneumatic mixing methods to maintain a constant oxygen volumetric mass transfer coefficient (K_La)[18–20,24], constant gas flow rate per unit volume (vvm)[18,20,23], or a combination of these criteria. To leverage these criteria and expedite process design and optimisation across various bioreactor scales and configurations, industries and academia commonly employ Computational Fluid Dynamics (CFD) as a powerful numerical tool to visualise bioreactor fluid flow and (bio) reaction-transport.

Depending on the operational Reynolds numbers and corresponding flow regimes of the bioreactor, CFD codes like ANSYS Fluent, used in this

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https://doi.org/10.1016/j.bej.2024.109585
Received 24 September 2024; Received in revised form 16 November 2024; Accepted 18 November 2024
Available online 19 November 2024

study, offer a variety of viscous models within their libraries. These commonly include turbulent regime simulations using k–ε [4,13,25], and k – ω [25–27] models, as well as transition regime simulations with SST models [5,6,28–33]. Irrespective of the chosen viscous model, only a handful of literature studies have integrated CFD-predicted flow fields with (bio)reaction kinetics for unified biosystem simulation. This integration has been achieved either within the CFD solver itself [7,14, 34–37], or by using multi-party software (i.e., CFD solver in conjunction with second-party software, such as MATLAB [8,38,39], Maple [3], and Paraview [40], or CFD solver alongside second and third-party software, such as OpenFOAM [41,42]). Compared to the latter approach, the former approach is reported to be numerically stable, building on its two-step integration: i.e., solving hydrodynamic to convergence in step one and freezing flow fields in step one to solve for bioreaction transport in step two. Additionally, the approach provides excellent computational cost savings when exploiting parallel computing [34,35] and using larger time sizes in step two to further expedite simulations.

However, these investigations were primarily focused on laminar [35,43] and turbulent [3,8,37,41] regimes, with a limited exploration into the hydrodynamics of transitional regimes, despite its relevance to scale-up/down of various benchtop, pilot, and industrial bioreactors like photobioreactor. In fact, the integration of transitional hydrodynamics regime models with photo-(bio)reaction kinetics has never been achieved to the best of the author's knowledge. While it may seem intuitive to apply the two-step approach for such mechanical mixing methods experiencing transitional hydrodynamics regime, it was found to be incompatible without substantially modifying the proposed methodology. This is because the authors [5,6] reported that the bioreactor's rotating stirrer generates unsteady radial flow, with each flow parcel deflecting upwards towards the liquid surface upon impingement on the bioreactor walls, and then returning towards the stirrer shaft. This flow pattern creates a recirculation zone and secondary vortices above the stirrer, leading to consistent flow fluctuations from all directions, which impedes the attainment of steady-state hydrodynamic convergence for step one. Consequently, proceeding to step two of the coupling with (bio)reaction kinetics becomes computationally impossible.

This computational intractability significantly slows the development and optimisation of bioprocesses, especially in high-throughput systems utilising miniaturised stirred bioreactors, such as magnetic stirrer-driven bioreactors. These reactors are widely used for strain screening [44,45], growth media and cultivating variable design [46–48], and operational strategy optimisation [49,50] due to their simplicity and low cost. Such systems have spurred numerous CFD investigations [5,6,47,51] aimed at improving the understanding of key parameters like mass transfer, heat transfer, and fluid dynamics, all of which are crucial for successful scale-up. However, the lack of integration between these CFD studies and (bio)reaction kinetics leads to suboptimal designs and inconsistencies during scale-up. This highlights an urgent need for more efficient coupling approaches and algorithms that can streamline fluid dynamics with (bio)reaction kinetics, enabling higher accuracy and reproducibility across scales.

On the one hand, time-averaging techniques have been used to address unsteady/oscillating flow fluctuations, resulting in smoothed steady flow fields in the literature. For example, López-Rosales et. al., [15] time-averaged gas holdup, liquid velocity, and local energy dissipation rate in a bubble column reactor over 30 seconds with a data sampling interval of 0.01 seconds. Similarly, Nauha et. al., [14] time-averaged all required flow variables in their compartmentalization model, including pressure, velocity, and turbulence dissipation, over 100 seconds using user-defined functions and storing intermediate results in memory locations within ANSYS Fluent cells. As well, Brown et. al., [33] time-averaged the transient flow variables over 150 seconds in their simulation of an industrial crystalliser, utilising various turbulence models, including *k*-ε, RNG *k*-ε, SST, SSG Reynolds Stress and Explicit Algebraic Reynolds Stress models (EARSM). While the SST model effectively captured the fluctuating velocity fields in the internal flow problem necessary for time-averaging, it showed significant deviations from the Laser Doppler Velocimetry data. This discrepancy was linked to inaccurate switching of the SST blending functions at the high Reynolds number of 300,000. Tamburini et al., [52] further supported these findings by comparing SST model to DNS simulations, and literature data across a Reynolds number range of 500–33,000. They observed that while the SST model showed considerable agreement in predicting power numbers for unbaffled and baffled vessels stirred by a Rushton turbine at lower Re $\leq$ 10,000, it exhibited increasing deviations beyond this threshold, aligning with the observations made by Brown et al. [33] at higher Reynolds numbers. These accurate simulations at low to midrange Reynolds number in internal flow problems are indicative of transitional regimes, making the SST model particularly well-suited for capturing the fluctuating velocity fields, as demonstrated in the referenced publications [5,6,32,52].

Despite these achievements, the time-averaging of oscillating flow variables from the bioreactor's rotating stirrer to hydrodynamic convergence in step one and subsequent coupling to bioreaction kinetics in step two has not been reported building around the similar issue of computational expense for a potential scale-up model. Looking at photobioreactors specifically, light transport models tailored towards utilising the Beer-Lambert law to reduce computational costs within CFD solvers have been primarily limited to rectangular geometries, such as flat plate [34,43,53], and simplified unidirectional 2D models for cylindrical geometries [35]. However, turbulence simulations in bioreactor are inherently 3D in nature, and resources for light transport models in 3D cylindrical geometries are scarce, complicating the second-step coupling of bioreaction kinetics. Although Mingjie et al., [37] recently introduced a light transfer model for uniformly illuminated 3D stirred tank bioreactor, similar to the works by Alfano et al. and Roger et al. [54,55], their approach requires substantial modifications for application to unidirectionally illuminated 3D cylindrical geometries. Furthermore, their second-step coupling of bioreaction kinetics was achieved using Lagrangian particle tracking, which overlooks spatial turbulent contributions by not solving scalar transport equation, an important factor of flows in transitional regimes.

Therefore, this manuscript aims to develop a 3D CFD model of a miniaturised cylindrical bioreactor, internally mixed by a rotating stirrer that induces oscillating flow fields, necessitating time-averaging for hydrodynamic convergence in the first step. This magnetic stirrer-driven bioreactor operates under varying Reynolds numbers, including the transitional flow regime, simulated with *transitional* SST turbulence models [29], and undergoes photo-biochemical conversions. This requires light transport models for the 3D CFD solver to couple photo-bioreaction transport in the second step. An exemplary literature case, the magnetic stirrer-driven Schott bottle photobioreactor (see [56]), often utilised for high-throughput setups was investigated herein. Additionally, the CFD-predicted flow fields are firstly validated with trace dye experiments and thereafter validated by simulating the photo-production of biomass by the photosynthetic bacterium *Rhodopseudomonas palustris* NCIMB 11774 (*R. palustris's*).

2. Methodology

2.1. Materials, experimental setup, and tracer conductivity measurements

A detailed description of the Schott bottle photobioreactor mixed by a magnetic stirrer can be found in our previous works [48,56,57]. The bioreactor vessel is made of glass for optimal light transparency and cylindrical in shape to minimised near-wall dead zones. Its overhead section was fitted with autoclavable GL45 polypropylene lids, gastight stainless-steel liquid sampling tubes, and the entire vessel submerged within a water bath for precise temperature control. However, the overhead section and water bath were omitted in this study to reduce the computation domain and minimise simulation costs. Therefore, only the adaptation of the experimental setup for tracer conductivity measurements is presented in Fig. 1 A). Two probes were used, submerged at 62.5 mm and 47.5 mm from the top of the bioreactor to detect tracer (i.e., NaCl) concentrations. Deionised water was poured into the bioreactor to a liquid height of 92 mm, corresponding to a working volume of 680 mL, and operated within stirring speeds ranging from 100 to 500 rpm, generating transitional and early turbulent flow regimes. Subsequently, 1 mL of 0.2 M NaCl was added into the bioreactor at time t = 0, and tracer concentration data were collected using the conductivity probe meters. The collected data was normalized and used for further analysis. NaCl (99 % purity) was purchased from Sigma Aldrich, and the numerical simulations considered the properties of water as follows: a density of 998.00 kg/m^3 and a viscosity of 1.00 mPa.s.

2.2. Photobioreaction kinetic model

The mechanistic model used to simulate *R. palustris's* photoheterotropical biomass growth was previously detailed in our publication [56] and is repeated in Eq. (1). As the substrate, glycerol concentration was considered replete (i.e., > 20 mM at the end of each batch [56]), it was omitted from this simulation to focus on the limiting effect of Light/dark (L/D) cycles generated by the magnetic stirrer's fluid dynamics and photobioreactor light attenuation. Intense culture mixing at higher rpms induces shorter L/D cycles, reducing cells' residence time in both the photic zone near the irradiated photobioreactor surface and the dark zone in the photobioreactor interior. This prevents photosynthetic proteins from becoming over-reduced or over-oxidised, benefiting growth dynamics of photosynthetic microbial species [34, 58–60].

$$\frac{dX}{dt} = \mu_B \bullet X = A' \bullet \exp\left(-\frac{E_a}{RT}\right) \bullet \left(\frac{\frac{I_0}{(\tau \bullet X) \bullet L} \cdot \left(1 - e^{-(\tau \bullet X) \bullet l}\right)}{k_s + \left(\frac{I_0}{(\tau \bullet X) \bullet L} \cdot \left(1 - e^{-(\tau \bullet X) \bullet l}\right)\right)}\right) \bullet X \tag{1}$$

Here, μ_B is the specific growth rate term (h^{-1}), X represents biomass concentration and t denotes photofermentation time, A' (8.30×10^3 h^{-1}) stands for the pre-exponential factors for biomass growth, E_a (2.74×10^4 J mol^{-1}) represents the activation energies for biomass growth, R (8.3145 J mol^{-1}K^{-1}) is the universal gas constant, T (308.15 K) denotes the absolute temperature, I_0 (100 Wm^{-2}) signifies the incident light intensity, k_s (500.0 Wm^{-2}) represents the light saturation coefficients for biomass growth, τ (90.8 m^2 g^{-1}) indicates the light absorption coefficient, L (0.098 m) signifies the light path length.

2.3. Computational Fluid Dynamics (CFD) modelling

2.3.1. Hydrodynamic model setup

The two phases simulated were a liquid bio-suspension and a gaseous headspace, both modelled as distinct phases using the Volume of Fluid (VOF) method [5,6,37,61]. The VOF method firstly solves a single set of continuity (i.e., Eq. (2.1)) and momentum (i.e., Eq. (2.2)) equations for the entire computational domain, as shown in Fig. 1 B) and C), following the methodology depicted in Fig. 2. Subsequently, the volume fraction of the phases, denoted as α_k, are incorporated into the computation of phase-dependent fluid properties such as density (i.e., Eq. (2.4)) and viscosity (i.e., Eq. (2.5)). The sum of the volume fractions equals unity (i.e., $\alpha_1 + \alpha_2 = 1$); therefore, $\alpha = 1$ indicates that the entire computational cell volume is occupied by the liquid bio-suspension phase and vice

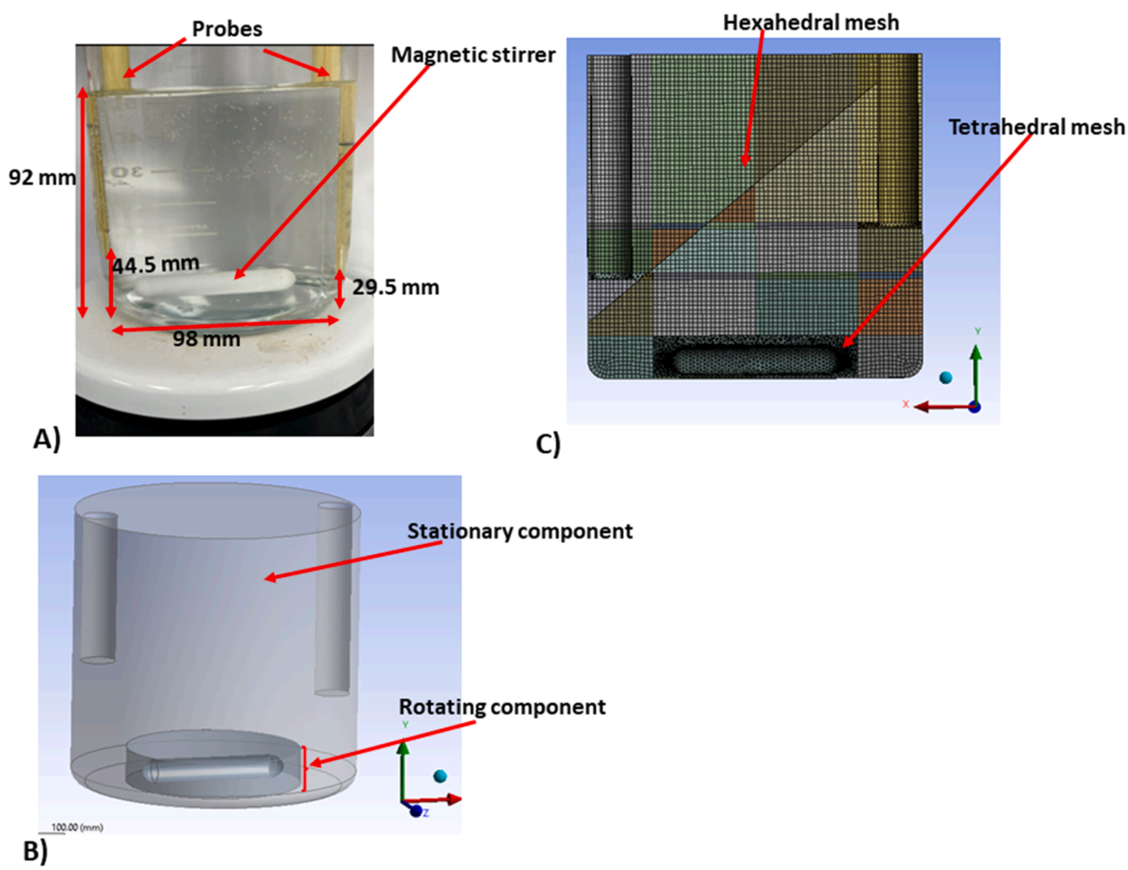

Fig. 1. Experimental and numerical setup: A) Schott bottle bioreactor mixed by magnetic stirrer, indicating bioreactor dimensions and probe inserts. B) 3D CFD geometry created for a multi-reference simulation approach featuring an inner rotating component with the magnetic stirrer and an outer stationary component representing the rest of the bioreactor, and C) hybrid mesh of hexahedral and tetrahedral elements utilised for the computational domain.

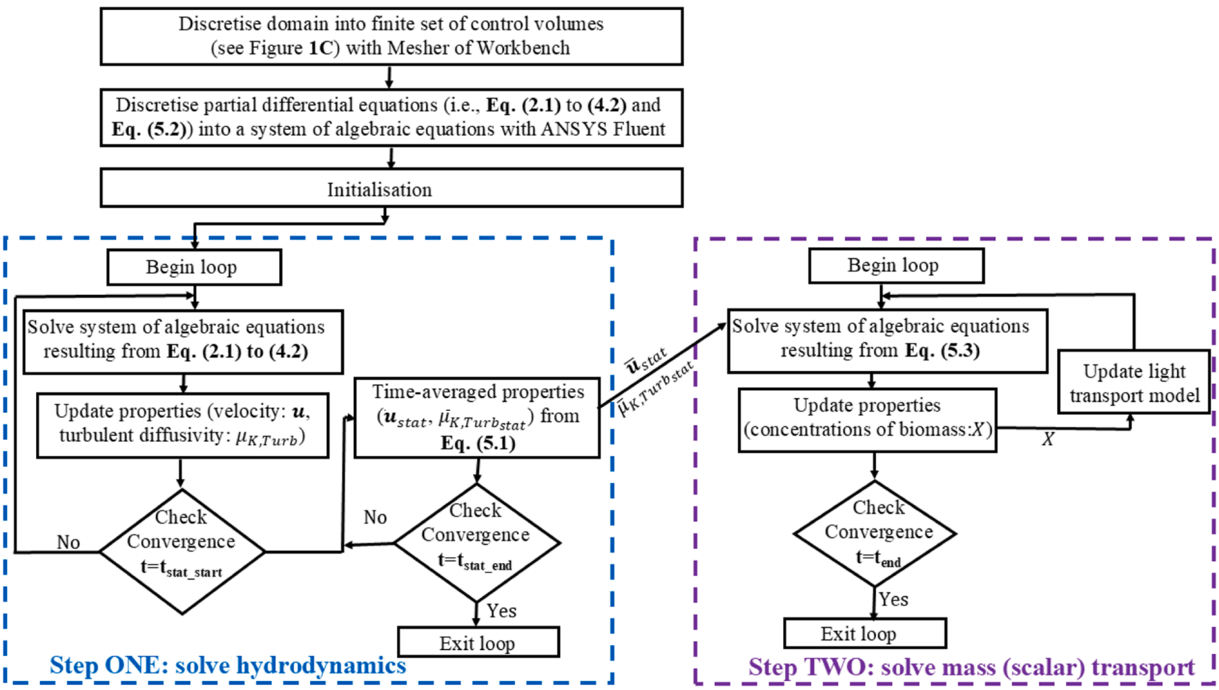

Fig. 2. A schematic illustrating the CFD coupling of time-averaged hydrodynamics to light transfer and biomass transport within a Schott bottle photobioreactor. The time-averaging triggers, t_{stat_start} and t_{stat_end} were set to 10 s and 50 s, respectively.

versa for $\alpha = 0$. The interface between the liquid bio-suspension and a gaseous phase (i.e., $0 < \alpha < 1$) was tracked using a conservation equation (Eq. (2.6)) for the interface mass balances.

$$\frac{\partial}{\partial t}(\rho) + \nabla \bullet (\rho \boldsymbol{u}) = 0 \tag{2.1}$$

$$\frac{\partial}{\partial t}(\rho \boldsymbol{u}) + \nabla \bullet (\rho \boldsymbol{u}\boldsymbol{u}) = -\nabla p + \nabla \bullet \left(\mu\left[\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T - \frac{2}{3}\mathrm{I}(\nabla \bullet \boldsymbol{u})\right]\right) + \rho \boldsymbol{F}_{\overrightarrow{\omega}k} + \rho \boldsymbol{g} + \boldsymbol{F}_{st} \tag{2.2}$$

$$\boldsymbol{F}_{\overrightarrow{\omega}k} = -2(\overrightarrow{\omega} \times \boldsymbol{u}) - [\overrightarrow{\omega} \times (\overrightarrow{\omega} \times \overrightarrow{r})] \tag{2.3}$$

$$\rho = \sum_{k=1}^{2} \alpha_k \rho_k \tag{2.4}$$

$$\mu = \sum_{k=1}^{2} \alpha_k \mu_{k,eff} = \sum_{k=1}^{2} \alpha_k (\mu_{k,Lam} + \mu_{k,Turb}) \tag{2.5}$$

$$\frac{\partial}{\partial t}(\alpha) + \boldsymbol{u} \bullet \nabla \alpha = 0 \tag{2.6}$$

Here, the subscript $k = 1, 2$ denotes the liquid bio-suspension and gaseous phases for the multiphase simulation. The terms, α_k, ρ_k, $\mu_{k,eff}$, $\mu_{k,Lam}$ and $\mu_{k,Turb}$ correspond to the phase volume fraction, density, effective viscosity, laminar and turbulent viscosities respectively. The symbol $\boldsymbol{u}$ represents the mean velocity in the context of the Reynolds-Averaged Navier-Stokes (RANS) modelling approach [62]. RANS was employed in this study due to its cost -effectiveness in simulation. Additionally, the terms p, I, $\boldsymbol{F}_{\overrightarrow{\omega}k}$, $\boldsymbol{g}$ and $\boldsymbol{F}_{st}$ refer to the shared pressure field, identity tensor, rigid body rotational body force of the magnetic stirrer, gravitational acceleration vector and surface tension forces respectively. The rotating magnetic stirrer induces two forces: Coriolis forces (represented by the first term of Eq. (2.3)) and centrifugal forces (represented by the second term of Eq. (2.3)). Here, the symbols, $\overrightarrow{\omega}$ and $\overrightarrow{r}$ represent the angular velocity of the rotating reference frame and the position vector.

This study explores several magnetic stirrer rotating speeds (100 rpm $< \overrightarrow{\omega} <$ 500 rpm), resulting in chaotic and stochastic movements characterised by visible eddies or swirls. This necessitates additional transport equations for closure of $\mu_{k,eff}$ in Eq. (2.5). For this purpose, the *four-equation transitional* SST model by F. R. Menter et al. [29], commonly called $\gamma - \mathrm{Re}_\theta$ model, was employed. The model integrates the SST $k-\omega$ model (i.e., Eqs. (3.1) to (3.2) including Table 1 for parameter definition and coefficients) [28,29] and two additional transport equations: one for intermittency (i.e., Eq. (4.1)) and the other for momentum thickness (i.e., Eq. (4.2)) to predict the transition onset and length. Firstly, it incorporates corrections for turbulent viscosity calculations (i.e., Eqs. 3.1–3.4 (c) in Table 1) from the turbulent shear stress in the near-wall region by blending the standard $k-\varepsilon$ and $k-\omega$ models using the functions F_1, and F_2, resulting in low-Reynolds-number corrections [28,63] in stagnation regions. Additionally, and most importantly, it embeds transition models into the SST $k-\omega$ model with Eqs. 3.3 to 3.5 by modelling intermittency to modulate the production of turbulence, thus triggering local transition effects. Note that when intermittency is close to zero (i.e., $\gamma \sim 0$), the turbulence production term vanishes (i.e., $\widetilde{P_k} \sim 0$ in Eq. (3.3)), causing the SST $k-\omega$ model to behave like a laminar model, meanwhile it becomes fully turbulent when $\gamma \sim 1$. For values $0 < \gamma < 1$, characteristic of flows in the transitional regime, the *four-equation transitional* SST model consistently outperforms other models found in literature [28,64]. Also, non-local transition effects to the flow, such as freestream turbulence intensity and length scale, *Re*, pressure gradient, etc. are captured by modelling the momentum-thickness Reynolds number, $\widetilde{\mathrm{Re}}_{\theta t}$, with Eq. (4.2).

$$\frac{\partial}{\partial t}(\rho k) + \nabla \bullet (\rho k \boldsymbol{u}) = \nabla \bullet \left(\left(\rho \mu_{k,Lam} + \frac{\mu_{k,Turb}}{\sigma_K}\right) \bullet \nabla k\right) + \widetilde{P_k} - \widetilde{D_k} \tag{3.1}$$

Table 1
Model equations relating to the original SST $k-\omega$ model [28] as described in Eqs. 3.1 to 3.4.

Model Description	Equation Definition	Equation Number
Parameter affecting the effective diffusivity of k	$\sigma_K = \dfrac{1}{\dfrac{F_1}{\sigma_{k,1}} + \dfrac{(1-F_1)}{\sigma_{k,2}}}$	Eqs. 3.1–3.4 (a)
Parameter affecting the effective diffusivity of ω	$\sigma_\omega = \dfrac{1}{\dfrac{F_1}{\sigma_{\omega,1}} + \dfrac{(1-F_1)}{\sigma_{\omega,2}}}$	Eqs. 3.1–3.4 (b)
Turbulent viscosity	$\mu_{k,Turb} = \rho \bullet \dfrac{k^2}{\omega} \bullet \dfrac{F_1}{\max\left[\dfrac{1}{\alpha^*}, \dfrac{SF_2}{\alpha_1 \omega}\right]}$	Eqs. 3.1–3.4 (c)
Production of k due to mean velocity gradients	$P_{k,} = \min[P_k, 10\rho\beta^* k\omega]$	Eqs. 3.1–3.4 (d)
Production of ω	$P_\omega = \dfrac{\rho\propto}{\mu_{k,Turb}} \bullet P_k$	Eqs. 3.1–3.4 (e)
Dissipation of k	$D_k = \rho\beta^* k\omega$	Eqs. 3.1–3.4 (f)
Dissipation of ω	$D_\omega = \rho\beta k\omega^2$	Eqs. 3.1–3.4 (g)
Cross diffusion term	$Cd_\omega = 2(1-F_1)\rho\sigma_{\omega,2}\dfrac{1}{\omega}\dfrac{\partial k}{\partial x_i}\dfrac{\partial \omega}{\partial x_j}$	Eqs. 3.1–3.4 (h)

Herein, the SST $k-\omega$ model constants (i.e., $\sigma_{k,1}$, $\sigma_{k,2}$, $\sigma_{\omega,1}$, $\sigma_{\omega,2}$, α^*, β^*, $\propto$, β) are often kept at their default values as specified in the ANSYS Fluent 2024R1 solver. This is supported by the fact that these coefficients have been calibrated for classical boundary layer flows. However, they can also be conveniently fine-tuned for specific applications. S is the strain rate magnitude, while F_1, and F_2, are blending functions that adjust the turbulent viscosity calculations.

$$\frac{\partial}{\partial t}(\rho\omega) + \nabla \bullet (\rho \boldsymbol{u}\omega) = \nabla \bullet \left(\left(\rho\mu_{k,Lam} + \frac{\mu_{k,Turb}}{\sigma_\omega} \right) \bullet \nabla\omega \right) + P_\omega - D_\omega + Cd_\omega \tag{3.2}$$

$$\widetilde{P_k} = \gamma_{eff} P_k \tag{3.3}$$

$$\widetilde{D_k} = \min\left[\max\left(\gamma_{eff}, 0.1\right), 1.0\right] D_k \tag{3.4}$$

$$F_1 = \max\left(F_{1orig}, F_3\right) \tag{3.5}$$

Here, k and ω represents the turbulent kinetic energy and specific dissipation rate respectively. The symbols P_k and P_ω denote the production of k and ω, while D_k and D_ω are the corresponding destruction terms. The parameters σ_K and σ_ω controls the turbulent viscosities, and Cd_ω is the cross-diffusion term from the original SST $k-\omega$ model [28,29] as reported in Table 1. The symbols $\widetilde{P_k}$, $\widetilde{D_k}$ and F_1 indicate corrections for turbulent production, destruction and the blending function that switches $k-\varepsilon$ and $k-\omega$ models. F_{1orig} is the original blending function of the SST $k-\omega$ model as reported in Tables 1 and 2. The term γ_{eff} is the effective intermittency obtained from Eqs. 4.1–4.2 (e) in Table 2 by solving Eqs. (4.1) to (4.2).

$$\frac{\partial}{\partial t}(\rho\gamma) + \nabla \bullet (\rho\gamma \boldsymbol{u}) = \nabla \bullet \left(\left(\mu_{k,Lam} + \frac{\mu_{k,Turb}}{\sigma_\gamma} \right) \bullet \nabla\gamma \right) + P_{\gamma 1} - E_{\gamma 1} + P_{\gamma 2} - E_{\gamma 2} \tag{4.1}$$

$$\frac{\partial}{\partial t}\left(\rho\widetilde{\mathrm{Re}}_{\theta t}\right) + \nabla \bullet \left(\rho \boldsymbol{u} \widetilde{\mathrm{Re}}_{\theta t}\right) = \nabla \bullet \left(\sigma_{\theta t}\left(\mu_{k,Lam} + \mu_{k,Turb}\right) \bullet \nabla \widetilde{\mathrm{Re}}_{\theta t} \right) + P_{\theta t} \tag{4.2}$$

Here, γ and $\widetilde{\mathrm{Re}}_{\theta t}$ denotes the intermittency and transition momentum-thickness Reynolds number respectively. The terms σ_γ and $\sigma_{\theta t} = 10$ [28,29] controls the diffusion coefficients. The terms $P_{\gamma 1}$ and $E_{\gamma 1}$ are the laminar-turbulent transition source terms whereas $P_{\gamma 2}$ and $E_{\gamma 2}$ are the destruction and re-laminarisation source terms, respectively. The term $P_{\theta t}$ is designed to force $\widetilde{\mathrm{Re}}_{\theta t}$ to match the local value of $\widetilde{\mathrm{Re}}_{\theta t}$ derived from empirical correlations outside the boundary layer. Refer to Table 2

Table 2
Model equations relating to the four-equation transitional SST model by F. R. Menter et al. [29]as described in Eqs. 4.1–4.2.

Model Description	Equation Definition	Equation Number
Transition sources	$P_{\gamma 1} = 2F_{length}\rho S[\gamma F_{onset}]^{c_{\gamma 3}}$ $E_{\gamma 1} = P_{\gamma 1}\gamma$	Eqs. 4.1–4.2 (a)
Destruction/relaminarisation sources	$P_{\gamma 2} = (2c_{\gamma 1})\rho\Omega\gamma F_{turb}$ $E_{\gamma 2} = c_{\gamma 2}P_{\gamma 2}\gamma$	Eqs. 4.1–4.2 (b)
Transition onset functions	$\mathrm{Re}_v = \dfrac{\rho y^2 S}{\mu}$ $R_T = \dfrac{\rho k}{\mu\omega}$ $F_{onset1} = \dfrac{\mathrm{Re}_v}{2.193\mathrm{Re}_{\theta c}}$ $F_{onset2} = \min\left(\max\left(F_{onset1}, F_{onset1}^4\right), 2.0\right)$ $F_{onset3} = \max\left(1 - \left(\dfrac{R_T}{2.5}\right)^3, 0.0\right)$ $F_{onset} = \max(F_{onset2} - F_{onset3}, 0.0)$ $F_{turb} = \exp-\left(\dfrac{R_T}{4.0}\right)^{4.0}$	Eqs. 4.1–4.2 (c)
Separation induced transition	$\gamma_{sep} = \min\left(2\max\left(\dfrac{\mathrm{Re}_v}{3.235\mathrm{Re}_{\theta c}} - 1, 0.0\right)F_{reattch}, 2.0\right)F_{\theta t}$ $F_{reattch} = \exp-\left(\dfrac{R_T}{20.0}\right)^{4.0}$ $\gamma_{eff} = \max\left(\gamma, \gamma_{sep}\right)$	Eqs. 4.1–4.2 (d)
Transition momentum thickness source terms	$P_{\theta t} = c_{\theta t}\dfrac{\rho}{t}\left(\mathrm{Re}_{\theta t} - \widetilde{\mathrm{Re}}_{\theta t}\right)(1.0 - F_{\theta t})$ $t = \dfrac{500\mu}{\rho U^2}$ $F_{\theta t} = \min\left(\max\left(F_{wake}\exp\left(-\dfrac{y}{\delta}\right)^{4.0}, 1.0 - \left(\dfrac{\gamma - 1/50}{1.0 - 1/50}\right)^{2.0}\right), 1.0\right)$ $\theta_{BL} = \dfrac{\widetilde{\mathrm{Re}}_{\theta t}\mu}{\rho U}$ $\delta_{BL} = \dfrac{15}{2}\theta_{BL}$ $\delta = \dfrac{500\Omega\gamma}{U}\delta_{BL}$ $\mathrm{Re}_\omega = \dfrac{\rho\omega y^2}{\mu}$ $F_{wake} = \exp-\left(\dfrac{\mathrm{Re}_\omega}{1E+5}\right)^{2.0}$	Eqs. 4.1–4.2 (e)
Coupling the transition model and SST transport Equations	$R_y = \dfrac{\rho y\sqrt{k}}{\mu}$ $F_3 = \exp-\left(\dfrac{R_y}{120}\right)^{3.0}$	Eqs. 4.1–4.2 (f)

Herein, the *four-equation transitional* SST model constants (i.e., $c_{\gamma 1}$, $c_{\gamma 2}$, $c_{\gamma 3}$, σ_γ, $c_{\theta t}$, $\sigma_{\theta t}$) were kept at their default values as specified in the ANSYS Fluent 2024R1 solver and as reported in F. R. Menter et al. [29]. This is supported by the fact that these coefficients have been calibrated for classical boundary layer flows. However, they can also be conveniently fine-tuned for specific applications.

for the equations and descriptions related to the *four-equation transitional* SST model.

2.3.2. Coupling of CFD hydrodynamics and (photo)bioreaction

In this study, the widely adopted two-step approach, involving first achieving hydrodynamics convergence before coupling to bioreaction kinetics, was employed with time-averaged calculations, as illustrated in Fig. 2. To ensure hydrodynamic convergence in step one, all necessary flow variables for solving the tracer and biomass transport model in Eq. (5.2) and Eq. (5.3) respectively, primarily velocities in the x, y, and z directions and turbulent diffusivity, were time-averaged using calculations prescribed in Eq. (5.1). These calculations were implemented with User Defined Functions written in the C programming language and solutions stored in User Defined Memories within ANSYS Fluent.

$$\overline{C}_{stat} = \frac{\sum_1^n C_{current}}{n} \tag{5.1}$$

$$\frac{\partial}{\partial t} = (\phi_L) + \nabla()\overline{\mu}_{stats}\phi_L = \nabla\left((\mu_{k,Lam})\nabla\phi_L\right) \tag{5.2}$$

$$\frac{\partial}{\partial t} = (X) + \nabla\,()\overline{\mu}_{stats}X = \nabla\left(\left(\mu_{k,Lam} + \frac{\overline{\mu}_{K,Turb_{stat}}}{S_{c_T}}\right)\nabla X\right) + \left[\mu_B \cdot X = A' \exp\left(-\frac{E_a}{R\cdot T}\right)\right] \frac{I_0 \exp^{(\tau\cdot X\cdot l\,)}}{I_0 \exp^{(\tau\cdot X\cdot l\,)} + K_s} X \tag{5.3}$$

Here $C_{current}$ denotes the RANS cell value (i.e., x-velocity, y-velocity, z-velocity, and turbulent diffusivity), n represents the number of time steps for which the $\overline{C}_{stat}$ averaging was performed over. The symbol $\varnothing_L$ represents the scalar tracer whilst the terms and $\overline{\mu_{K,Turb_{stat}}}$ denotes the time averaged statistic of the velocity components, and turbulent diffusivity respectively. The laminar diffusivity $\mu_{k,Lam} = 5.5 \times 10^{-10}$ m^2 s^{-1} [34,35,62] and Schmidt number, Sc_t 0.7 [34,62] were obtained from literature.

On the left-hand side of Eq. (5.2) and Eq. (5.3), the first and second terms represent the accumulation and convection of the tracer and biomass species, respectively. These equations, by default, retain the instantaneous RANS solution values in the CFD solver. Consequently, the computed time-averaged velocities (i.e., x-velocity, y-velocity, z-velocity) from the UDF were patched into the solver to replace the instantaneous RANS solution at the final time step of stage one. On the right-hand side, the first term denotes diffusion of the tracer (i.e., for validating the modelling results) and biomass species while the second term corresponds to growth. For Eq. (5.2), this growth term is zero because a fixed concentration of 0.2 M NaCl in 1 mL was injected at the start with no production term. The biomass growth term of Eq. (5.3), incorporates a light transport model (i.e., $I_0 \bullet \exp^{(-\tau\bullet X\bullet l)}$) to be simulated in the 3D CFD framework, capturing the L/D cycles effects via the μ_B term. This was achieved with the algorithm outlined in Fig. 3, implemented using User Defined Functions written in C, compiled and inserted into the Fluent solver. This approach employs polar coordinates to orient the incident light position and account for the photobioreactor's curvature; cartesian coordinates to estimate the light path length from the incident light position; User Defined Memories to store calculated values; Beer-Lambert equation for light attenuation [34,35], similar to previous literatures by Alfano et al. and Roger et al. [54,55]. The model is solved at every iteration during the simulation.

2.3.3. CFD solver settings and solution strategy

For the first stage of hydrodynamic simulations, a small-time step size of 0.001 s was used, with 40 iterations per time step and a total simulation time of 50 s. Convergence was considered achieved when residuals dropped below 10^{-6} and changes in velocity from one iteration to the next were negligible at all monitoring points. Although time averaging was employed in this stage, it was only initiated after the initial 10 s of simulation, resulting in a 40 s time-averaged solution for the hydrodynamic simulations. In the second stage, where scalar transport was coupled to hydrodynamic simulations, a larger time step size of 0.01 s was used, maintaining the same number of iterations per time step as in stage one. The total simulation times were 150 s for the tracer dye simulations and 6000 s for the biomass transport simulations. To match the CFD simulation duration of 6000 s with the 8 days (i.e., 691,200 s) photo-fermentation period, the maximum specific growth

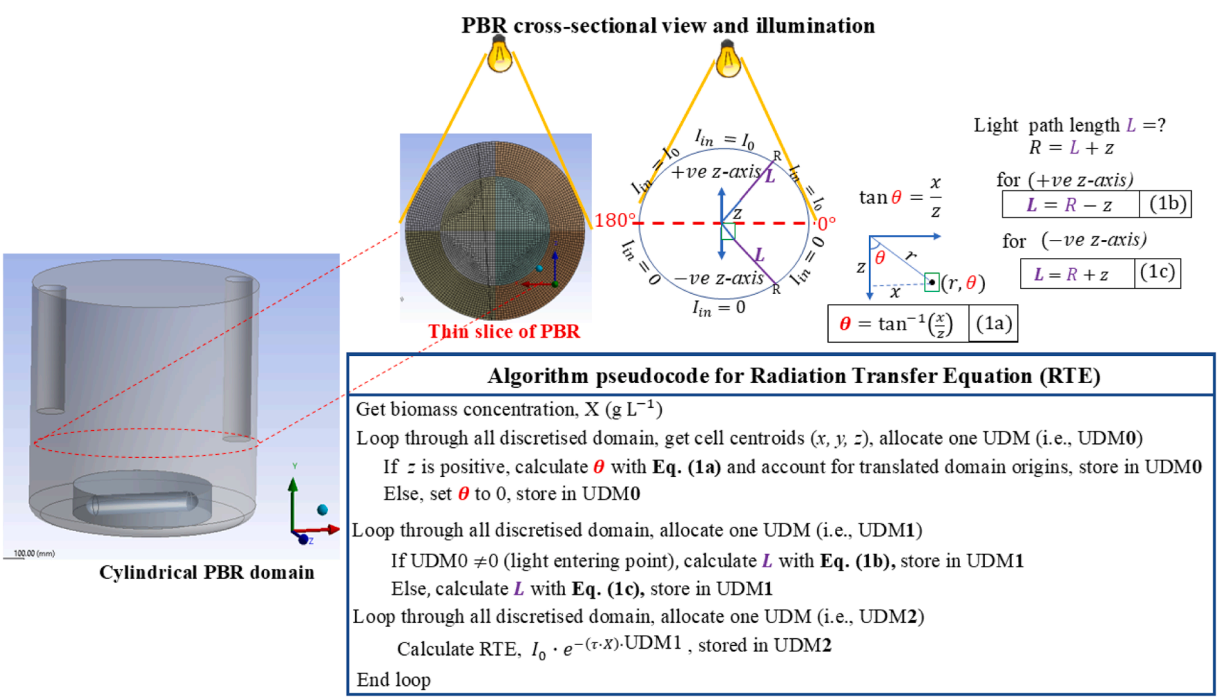

Fig. 3. Schematic illustration of unidirectional illuminated Schott bottle bioreactor alongside pseudocode for an algorithm that couples light intensity and light attenuation to scalar transport. Eq. (1a-c) are used to orientate the light entering point and calculate the path length. The Beer Lambert model common to other literatures [34,35,54,55] is then solved for light attenuation.

rate term in Eq. (5.2) (i.e., $A' \bullet \exp\left(-\frac{E_a}{R \bullet T}\right)$) was accelerated by a factor of $1.152 \bullet 10^3$. This scaling leverages the faster convergence of fluid dynamics (i.e., order of seconds [34,65]) compared to cell growth, which progresses over longer timescale (i.e., doubling time at an order of days [34,66,67]). By accelerating biomass growth kinetics up to an order of minutes or hours, the cells still experienced thousands of the light/dark cycles, closely approximating true biomass growth conditions, where cells experienced light/dark cycles over hundreds of thousands of times. This approach, as demonstrated in our previous works [34,35,62], significantly reduces computational costs without compromising solution accuracy. Further computational cost savings were achieved through parallel computing, utilising 16 and 32 processors for stage one hydrodynamic simulations to investigate time-averaging influences and 32 processors for stage two scalar transport simulations. Each processor core was allocated 4 GB of memory, and the implementation was carried out using the High-Performance Computing (HPC) cluster at The University of Manchester, United Kingdom.

3. Results and discussion

3.1. Grid sensitivity analysis, time averaging, and computational cost analysis

To evaluate the mesh independence of the CFD model prediction, three different meshes, namely Mesh1, Mesh2, and Mesh3, with element counts of 406,667, 267,460 and 151,291 respectively, were set up. The solution was monitored at three vertical lines, denoted L1, L2, and L3, corresponding to vertical heights of 23.0 mm, 46.0 mm, and 69.0 mm from the base of the bioreactor. As expected, increasing the mesh element count resulted in longer hydrodynamic simulation times, ranging from $1.5664 \bullet 10^4$ seconds (i.e., 4.351 hrs) for Mesh3 to $1.17766 \bullet 10^5$ seconds (i.e., 32.713 hrs) for Mesh1. However, the monitored vertical velocities in Fig. 4 A), and B) showed similar fluctuating behaviour across all monitoring positions within the bioreactor, rendering them mostly indistinguishable. These fluctuating flow fields arise from the unsteady radial flow generated by the bioreactor's rotating magnetic stirrer, a phenomenon also observed by other researchers [5,6]. However, these studies did not elaborate on the magnitude of velocity fluctuations, specifically the range of $-0.126\ \mathrm{m\ s^{-1}} \leq \boldsymbol{u}_y \leq 0.0\ \mathrm{m\ s^{-1}}$ and $0.0\ \mathrm{m\ s^{-1}} \leq |\boldsymbol{u}| \leq 0.211\ \mathrm{m\ s^{-1}}$, as seen in Fig. 4 for miniaturised stirred bioreactors. In contrast, Brown et. al., [33] reported velocity magnitude fluctuations between $0.0\ \mathrm{m\ s^{-1}} \leq |\boldsymbol{u}| \leq 0.42\ \mathrm{m\ s^{-1}}$ for their simulation of an industrial crystalliser, which aligns within the bounds observed in Fig. 4, supporting the validity of these results. This implies that further calculations dependent on these fluctuating fields will yield similarly varying output results, making the second step of coupling hydrodynamics and bioreactor kinetics intractable due to the lack of a unique solution.

Intuitively, Mesh2 was chosen for all subsequent calculations due to its balanced trade-off between accuracy and computational cost. Conversely, time-averaging the instantaneous RANS solution over an extended period, as observed in Fig. 4 C), D) and E), stabilises the flow fields, leading to converged steady states. This occurs due to negligible changes in velocities from one iteration and/or time step to the next over the moving horizon, enabling the second step of coupling hydrodynamics and bioreactor kinetics. Even so, the computational burden of time-averaging the instantaneous RANS solution remains unclear. Therefore, it was reasonable to evaluate the cost savings achieved by doubling the compute processors from 16 to 32 during the time-

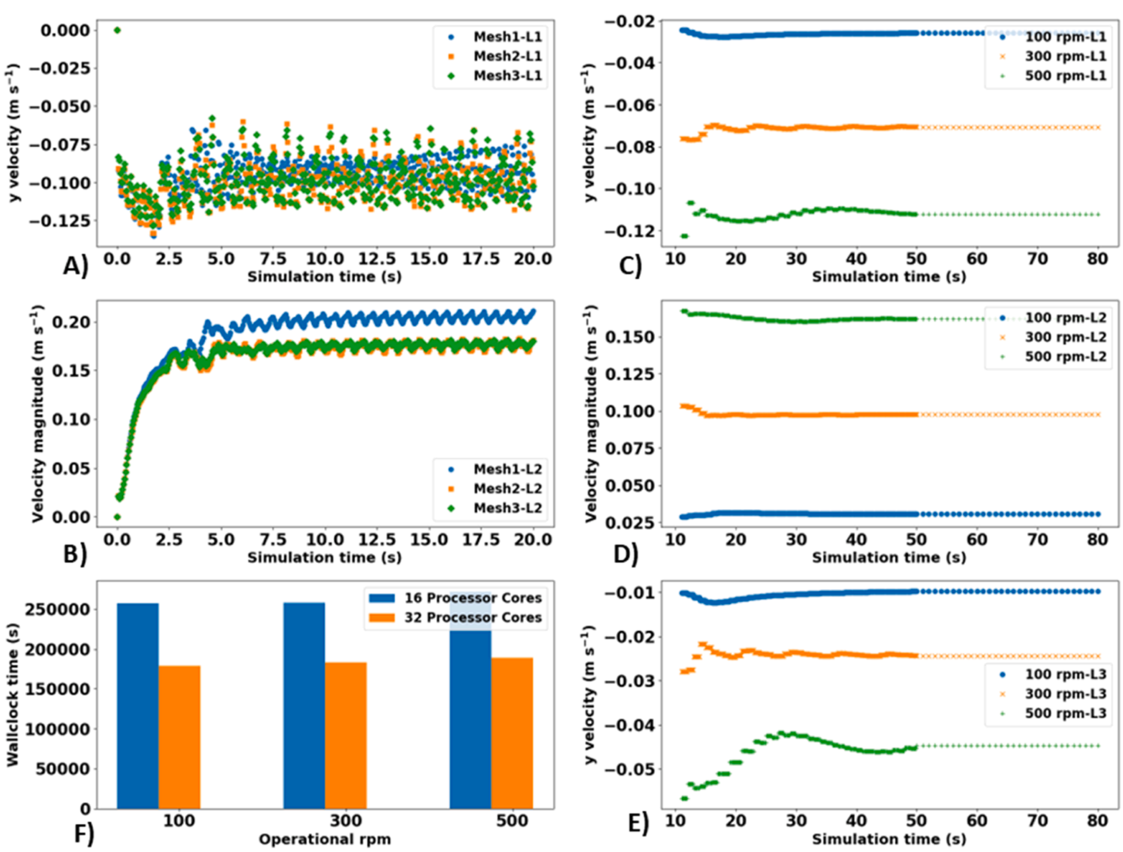

Fig. 4. CFD hydrodynamic RANS solutions for under different mesh element counts at vertical heights of A) 0.023 m, B) 0.046 m. Time-averaged RANS solutions at vertical heights of C) 0.023 m, D) 0.046 m, and E) 0.069 m, under varying operational stirrer rotational speeds. Additionally, F) illustrates the impact of increasing number of processor cores on simulation cost during the time-averaging.

averaging of the instantaneous RANS solution. Fig. 4 F) shows that the cost savings achieved for augmenting compute processors cores do not scale linearly with time-averaging, spanning all investigated stirrer rotational speeds, from 100 to 500 rpm. This suggests a notable influence on the overall computational expenses associated with stage one hydrodynamic convergence, potentially posing challenges for upscaled bioreactor simulations. These findings indicate the need for significantly more processor compute nodes (i.e., well beyond 32 cores) when simulating larger scale bioreactors. However, purchasing licenses for a large number of cores in commercial CFD software can be expensive. Therefore, more affordable open-source alternatives, such as OpenFOAM, are herein recommended as viable options. As well, cost-saving

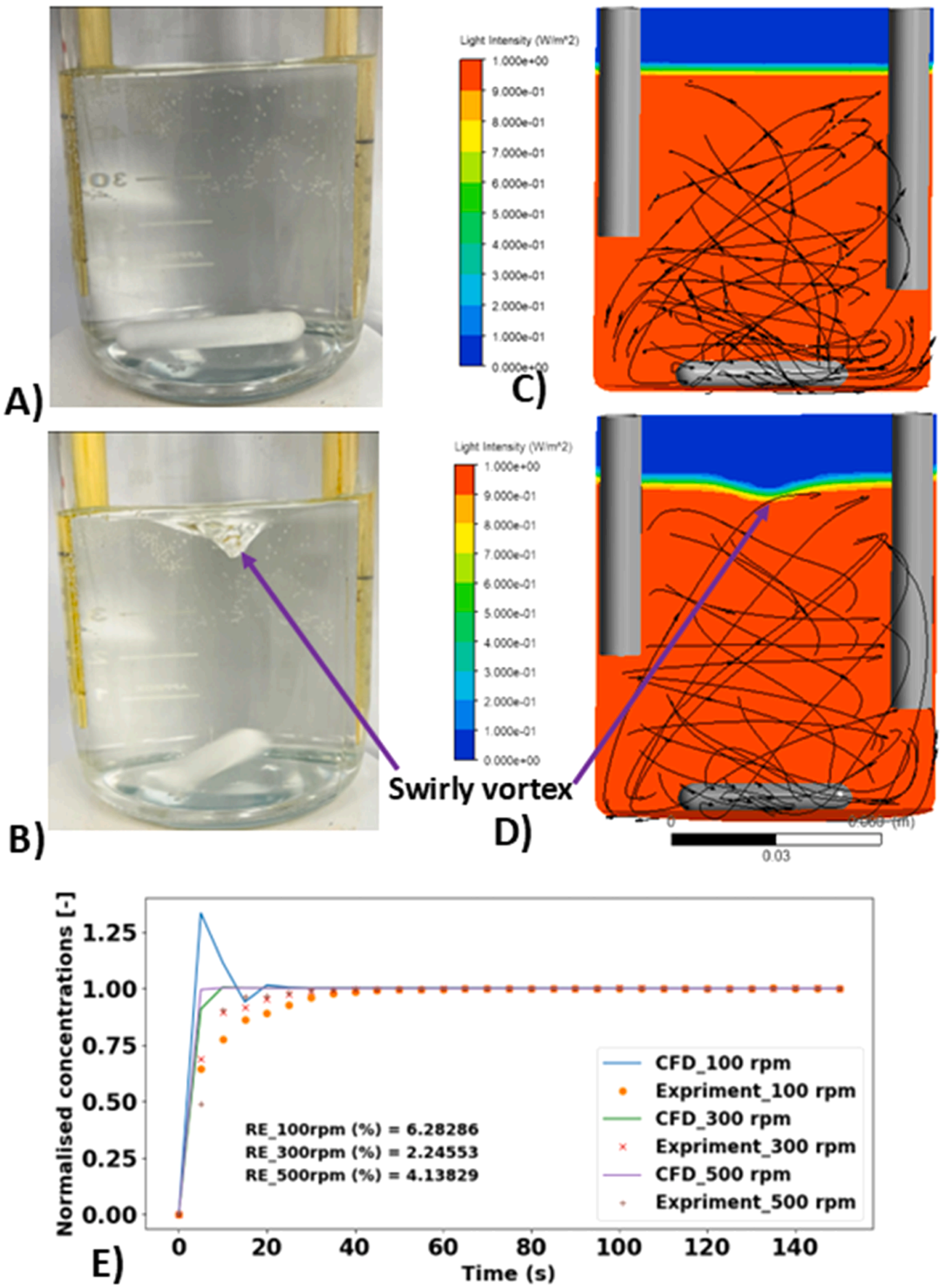

Fig. 5. Flow visualisation photographs of the physical bioreactor operating at A) 100 rpm, B) 500 rpm compared to the CFD predicted flow fields at C) 100 rpm, D) 500 rpm. The presences of swirly vortices in B) and D) confirm qualitative validations. Quantitative validation is presented in E), indicating the percentage relative error (RE) at various stirrer rotational speeds.

strategies like compartment modelling [14,68,69], which bypass the need to solve momentum balances and continuity equations at every element of a discretised geometry, offer a viable alternative for accelerating industrial-scale bioreactor simulations. However, the constantly fluctuating nature of the flow variables may necessitate a large number of compartments, which could undermine the intended cost savings. This is particularly true for the second-step coupling of bioreaction kinetics, where accurate light attenuation calculations are highly dependent on spatial discretisation.

3.2. CFD hydrodynamic insights and flow validations

To compare and validate the CFD model's predictions against experimental results, both qualitative and quantitative approaches were employed. Qualitatively, visual inspection was used to distinguish zones of liquid (water) within the bioreactor and air overhead, as shown in Fig. 5 (A – D), confirming the suitability of the VOF method in representing the multiphysics. Further qualitative analysis of the water zone, indicated by arrowheads in the flow fields in Fig. 5 C) and D), shows radial recirculation zones where fluid parcels impinge on the bioreactor walls and return towards the stirrer shaft in a swirling manner. These phenomena are consistent with the literature reports [5,6,52] but the observation becomes more pronounced with increasing stirrer rotational speeds, leading to the development of a swirling vortex resembling a tornado. This is especially evident at the high rotational speed of 500 rpm, as visualised in Fig. 5 B) and D), although its impact on biomass growth has not been discussed in other literature.

The quantitative analysis employed normalised conductivity probe measurements to compare against the CFD predictions when solving Eq. (5.2) on the stage one hydrodynamics fields. Fig. 5 E) shows a considerable agreement between the CFD predictions and the experimental measurements, with deviations within 6.3 %. This level of accuracy is acceptable for numerical validations, as other literature studies have reported deviations within 10 % [16,35,70,71]. Notably, the deviations were primarily observed within the first 16 seconds after introducing the tracer into the setup, with the model aligning closely with the equilibrium values beyond 20 seconds. This suggests that the model delivers accurate solutions after sufficient simulation time has elapsed. The initial discrepancy might be due to the CFD model's limitations in accurately representing the tracer's diffusivity upon its initial introduction into the computational domain. This explanation aligns with findings by Teke et al. [72], who reported similar inconsistencies during early phase validations in CFD simulations of 5 L stirred tank bioreactor using standard $k-\varepsilon$ turbulence model, unlike the SST model used here. Additionally, the lower rpm of 100 exhibited the largest overshoot, resulting in the greatest deviations, which suggests a decline in accuracy as rpm decreases. This could be attributed to the reduced mixing velocity recirculating the tracer at low rpm, which delays tracer detection and compromises the accuracy of the technique. This limitation is also observed in other probe-based tracer validation techniques [6,73–75]. For instance, Allonneau et al., [6] used an optical fibre probe detection system and showed similar overshoots around the equilibrium value in their CFD simulations. Therefore, for low rpm validations with fluctuating dynamics, it is recommended to use non-intrusive validation techniques, such as Laser Doppler Anemometry (LDA), which avoid the limitations associated with probe-based methods.

3.3. Hydrodynamics influence on biomass growth and light transport

With the stage one hydrodynamic simulations converged, as shown in Fig. 4 C) to E), for all stirrer rotational speeds, the stage two coupling to bioreaction simulations can now be analysed to assess the hydrodynamic effects on bioprocess performance. To elucidate these effects, Fig. 6 A) shows the expected increase in biomass production with higher stirrer rotational speeds: from 100 rpm to 300 rpm, and 500 rpm, yielding final biomass concentrations of 0.88 g/L, 1.66 g/L, and 1.68 g/L, respectively. The biomass growth profiles mimic those of purely mechanistic models in the literature [56], validating the rigorous coupling of biomass transport. These stage two simulations were executed with 32 processors, requiring $1.68313 \bullet 10^5$ seconds, $1.67022 \bullet 10^5$ seconds, and $1.63762 \bullet 10^5$ seconds of wall clock time for 100 rpm, 300 rpm, and 500 rpm, respectively. When combined with stage one, the total simulation times using 32 processor cores were $3.47006 \bullet 10^5$ seconds (~96.4 hours), $3.49786 \bullet 10^5$ seconds (~97.2 hours), and $3.52640 \bullet 10^5$ seconds (~98.0 hours), respectively. This amounts to approximately four days of wall clock time, indicating significant cost savings compared to the supposed several weeks or months of computation without using the accelerated growth kinetic approach earlier reported [34,35,62], meanwhile not adversely affecting simulation accuracy.

The model indicates an 88.16 % increase in biomass production when increasing the stirrer speed from 100 rpm to 300 rpm, compared

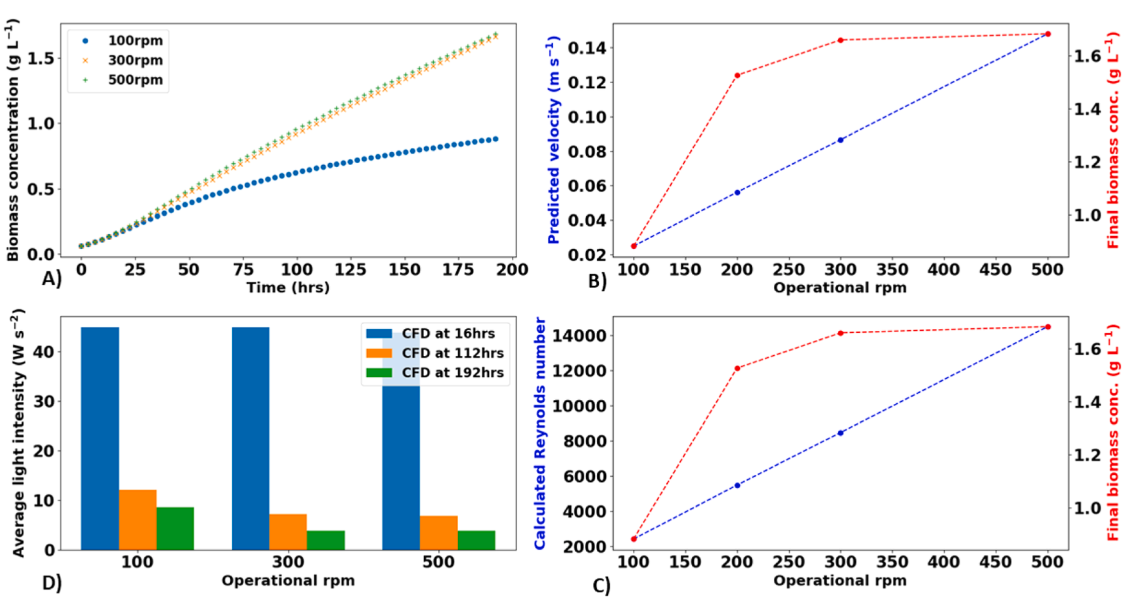

Fig. 6. CFD predictions for: A) temporal changes of biomass concentration over time; and the relationships between biomass changes and the bioreactor's operating parameters: B) volume-averaged velocity; C) calculated Reynolds number; and D) volume-averaged light intensity.

to a 1.4 % increase from 300 rpm to 500 rpm. Such increases in biomass productivity are often attributed to improved culture mixing [34,59,76], particularly in the radial direction [34,58,60], which enables more frequent light/dark cycles for microbial cells to absorb light energy for photosynthesis. Fig. 6 B) analyses the stirrer-induced mixing volume-averaged velocities, showing a linear increase from 100 rpm to 500 rpm, corroborating other literature reports for both baffled [4,6,77,78]and unbaffled bioreactors [5,52,79]. At higher stirrer rotational speeds beyond 300 rpm, the linear increase in mixing velocity did not significantly enhance biomass productivity. This can be attributed to excessively rapid light/dark (L/D) cycles, which reduce the relaxation time of excited photosystems and limit the utilisation of absorbed energy for growth [34], effectively keeping the microbial cells in almost continuous light. Such too fast L/D cycles have been reported in purely turbulent regimes in the literature [34,76]. This interpretation is supported by the Reynolds number results in Fig. 6 C), showing 14,481.60 at 500 rpm, nearly double that at 300 rpm (8469.21). Previous studies [48,57,80] and our work [56] have operated these bioreactors at 200 rpm but have not investigated beyond this 200 rpm benchmark.

Since the CFD model has been duly validated, its full potential was exploited beyond the 200 rpm benchmark to explore light transport limitations on biomass growth. Fig. 7 shows light attenuation during the bioprocess dynamics at 16 hours, 112 hours, and 192 hours for all operational rotational speeds. This is expected, as continuous cell growth requires intensified light absorption due to mutual shading, as observed elsewhere [58–60,81], thus confirming the reliability of the proposed algorithm in Fig. 3 for simulating light transport. Interestingly, light attenuation intensified with increasing rpm, becoming severe beyond 300 rpm. This suggests that microbial cells are exposed to insufficient light intensity at these higher rpms, as confirmed in Fig. 6 D), where the average light intensity was below 7.2 W/m^2 at 300 rpm and 6.9 W/m^2 at 500 rpm at 112 hours. At such low light-limiting levels, increasing the rpm from 300 to 500 rpm is expected to have a negligible influence on biomass productivity, as seen in Fig. 4. Conversely, augmenting the incident light intensity from 100 W/m^2 should provide more photo energy to the bioreactor, potentially enhancing biomass productivity at elevated rpms. This interpretation aligns with findings in the literature across various photobioreactor designs, including stirred tank reactors [82–85], shake flask reactors [86], cylindrical bubble columns [81,87–89], and flat panel reactors [34,76,90,91], all of which have been studied with different photosynthetic microorganisms such as microalgae and cyanobacteria.

3.4. Potential applicability of the proposed CFD model

The full potential of exploring previously untested bioreactor rotational speeds is demonstrated in Section 3.3. However, testing augmented incident light intensities remains pending, especially since our previous investigation with the same microbial strain [56] explored up to 200 W/m^2 without severe photoinhibition. As a caveat, the mechanistic model used herein did not include the biomass decay rate term, which is often incorporated into dynamic biokinetic models to account for microbial cell deaths due to all biotic and abiotic factors, such as hydrodynamic induced shear stresses [92]. This omission could potentially lead to an overestimation of biomass growth, as it does not account for detrimental effects of increasing shear stresses due to increased bioreactor rotational speeds and light stresses from augmenting incident light intensities in future studies. For example, the results in Fig. 6 A) did not show a stationary growth phase even after 200 hrs which is somewhat unrealistic, as cellular activities typically decline as the bioprocess ages. This explains the observed sigmoid-shaped growth curve, where all distinct growth phases— lag phase, primary growth phase, secondary growth phase, and stationary phase—are integrated into dynamic models that include a decay term [34,88,93]. Therefore, it is paramount to incorporate the decay rate term into the model before using it to optimise rotational speeds and light intensity.

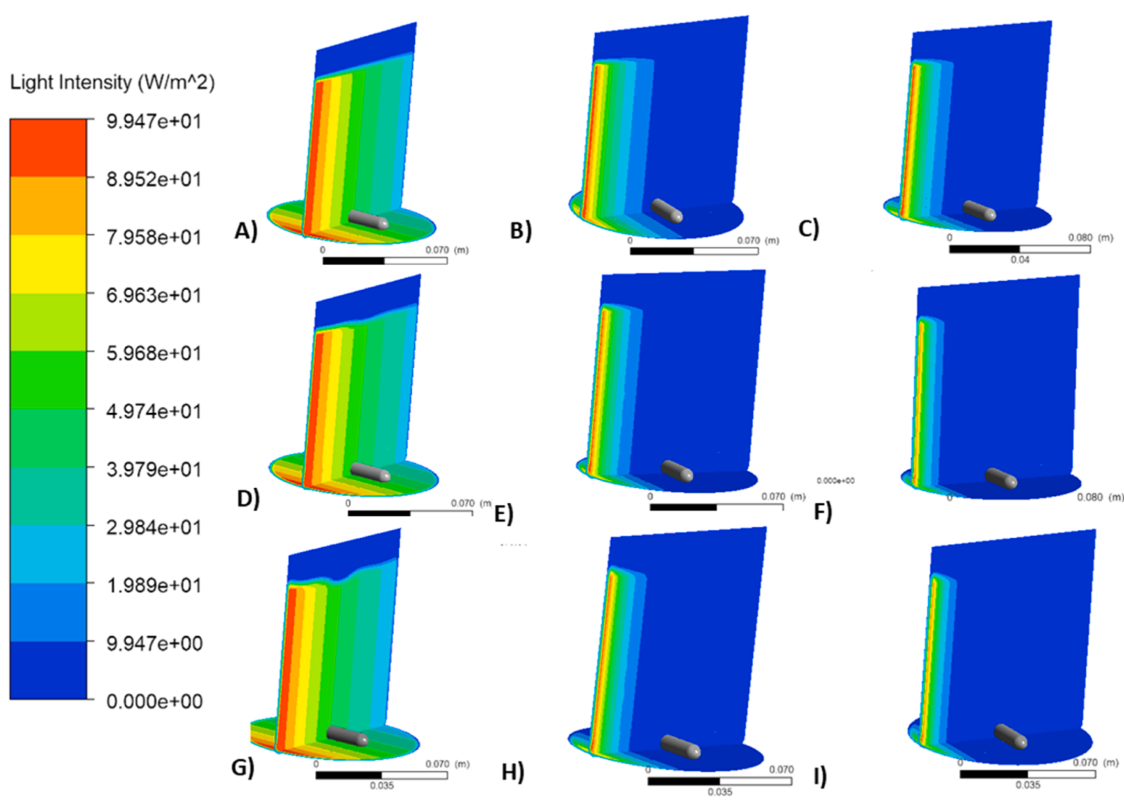

Fig. 7. CFD qualitative predictions of light transport dynamics in the photobioreactor operating various speeds, namely at 100 rpm: A) to C), at 300 rpm: D) to F), at 500 rpm: G) to I). These results were sampled at three-time interval levels at 16hrs: A), D) and G), 112 hrs: B), E) and F), and 192 hrs: C), F) and I). Severe light attenuation observed at higher operating speeds is corroborated by increased biomass growth.

Nonetheless, the proposed technique of time-averaging the instantaneous RANS solution before two-step coupling to bioreactor and light transport algorithms can be extended to model other bioreactors and photosystems.

4. Conclusions

Synergising hydrodynamics with bioreaction transport using a two-step approach becomes impractical for fluctuating hydrodynamic fields, which is the focus of this study. To address this, hydrodynamic convergence was ensured by time-averaging the instantaneous RANS solutions over a sufficiently long period to achieve statistical significance in step one. This enabled coupling to bioreaction transport on the time-averaged fields. Therefore, User Defined Functions (UDFs) were developed to replace the default instantaneous RANS fields within the CFD solver with the computed time-averaged instantaneous RANS fields for simulating bioreaction transport. To verify this approach, a Schott bottle photobioreactor driven by a magnetic stirrer, characterised by fluctuating radial velocities, was identified as an ideal candidate. This bioreactor setup was used to rigorously validate the proposed algorithms for tracer transport experimentally to within 7 % deviations and qualitatively for predicted swirly vortices fields. The time-average algorithm was observed not to scale linearly for the increasing number of processors, thus computationally intense.

Model predictions for light and biomass transport were validated against literature observations for stirrer rotational speeds below 200 rpm. While increasing stirrer rpm correlated with enhanced biomass productivity, unforeseen experimental scenarios beyond 200 rpm but less than 500 rpm were thoroughly scrutinised. Severe light attenuation and light limitation observed at higher operating speeds indicated the necessity of operating the photobioreactor at incident light intensities augmented above 100 W m^{-2}. Therefore, future efforts will focus on optimising the bioreactor's stirrer rotational speed and incident light intensities.

CRediT authorship contribution statement

Dongda Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology. **Robbert William McClelland Pott:** Writing – review & editing, Writing – original draft, Visualization, Resources. **Bovinille Anye Cho:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Godfrey K. Gakingo:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **George Mbella Teke:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Dr. Bovinille Anye Cho is a Royal Society Career Development Fellow, and gratefully acknowledges the funding provided by the Royal Society, United Kingdom.

Data Availability

Data will be made available on request.

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